

How Does Patient Capital Empower Supply Chains?

An Empirical Investigation of Spillover Effects from Government-Guided Funds

Xin Li¹, Xinlong Xiao^{2,*}

School of Finance, Capital University of Economics and Business (CUEB), Beijing, China

Abstract

This paper examines the impact of patient capital investment on alleviating the financing constraints of upstream and downstream firms in the supply chain. The findings reveal that patient capital investment generates a positive spillover effect within the supply chain, alleviating the financing constraints of core suppliers and customers. This spillover effect is more pronounced when the interdependency between supply chain firms is stronger and when the financing capacity of these key partners is weaker. Mechanism analysis indicates that patient capital operates through two channels: capital cost effect (improving accounts receivable turnover efficiency of suppliers) and credit expansion effect (increasing the proportion of credit loans for customers). Further analysis shows that this supply chain spillover effect significantly enhances the investment efficiency and innovation level of major suppliers and customers. The findings provide theoretical and empirical insights for policies aimed at fostering patient capital, promoting industrial and supply chain optimization and upgrading, and building a modern industrial system.

Keywords: Patient Capital; Supply Chain; Spillover Effect; Financing Constraints

¹ Xin Li, Ph.D. from Renmin University of China, Professor and PhD Supervisor, School of Finance, Capital University of Economics and Business. Research Interests: Capital Markets, Securities Investment. Email: lixin@cueb.edu.cn (X. Li)

² Xinlong Xiao (Corresponding Author), PhD Candidate, School of Finance, Capital University of Economics and Business. Research Interests: Corporate Finance, Capital Markets. Email: 12024110007@cueb.edu.cn (X. Xiao)

How Does Patient Capital Empower Supply Chains?

An Empirical Investigation of Spillover Effects from Government-Guided Funds

Xin Li¹, Xinlong Xiao^{2,*}

School of Finance, Capital University of Economics and Business (CUEB), Beijing, China

Abstract

This paper examines the impact of patient capital investment on alleviating the financing constraints of upstream and downstream firms in the supply chain. The findings reveal that patient capital investment generates a positive spillover effect within the supply chain, alleviating the financing constraints of core suppliers and customers. This spillover effect is more pronounced when the interdependency between supply chain firms is stronger and when the financing capacity of these key partners is weaker. Mechanism analysis indicates that patient capital operates through two channels: capital cost effect (improving accounts receivable turnover efficiency of suppliers) and credit expansion effect (increasing the proportion of credit loans for customers). Further analysis shows that this supply chain spillover effect significantly enhances the investment efficiency and innovation level of major suppliers and customers. The findings provide theoretical and empirical insights for policies aimed at fostering patient capital, promoting industrial and supply chain optimization and upgrading, and building a modern industrial system.

Keywords: Patient Capital; Supply Chain; Spillover Effect; Financing Constraints

¹ Xin Li, Professor and PhD Supervisor, School of Finance, Capital University of Economics and Business. Research Interests: Capital Markets, Securities Investment. Email: lixin@cueb.edu.cn (X. Li)

² Xinlong Xiao (Corresponding Author), PhD Candidate, School of Finance, Capital University of Economics and Business. Research Interests: Corporate Finance, Capital Markets. Email: 12024110007@cueb.edu.cn (X. Xiao)

1. Introduction

As supply chain collaboration models gradually replace traditional vertical integration to become the core organizational form of modern industrial systems, the overall competitiveness of supply chains has emerged as a key indicator for measuring a nation's economic resilience and development level. However, enterprises across the supply chain—particularly small and medium-sized businesses—universally face constraints from a lack of long-term financing sources during their development, creating a significant bottleneck that hinders the overall efficiency of industrial chains. Against this backdrop, patient capital—with its long-term orientation, risk tolerance, and deep engagement—offers new possibilities for addressing supply chain financing challenges. Therefore, examining the enabling mechanisms and transmission pathways of patient capital within supply chains holds significant theoretical and practical implications for enhancing the resilience and security of industrial and supply chains.

Existing theories have provided preliminary logical support for the potential spillover effects of patient capital within supply chains. Against the backdrop of structural limitations in traditional bank credit systems—such as maturity mismatches and risk aversion—patient capital demonstrates unique complementary financing functions through its long-term orientation and high risk tolerance (Zheng et al., 2000; Zhang et al., 2017). From a theoretical perspective, patient capital can reduce information asymmetry between enterprises and financial institutions through its “certification effect,” thereby improving credit allocation efficiency (Li and Ma, 2014). Simultaneously, its “governance effect” optimizes corporate financial decision-making and financing behavior (Xiong et al., 2020; Lei and Liu, 2017). These studies provide a theoretical foundation for understanding the transmission mechanisms of patient capital within supply chain networks, suggesting it may generate cross-firm spillover effects by alleviating financing constraints among node enterprises.

Against this backdrop, this paper utilizes data on the top five suppliers and customers disclosed by Chinese A-share listed companies, combined with patient capital investment information. By constructing a supply chain-capital linkage sample, it empirically examines the spillover effects of patient capital within supply chains and its underlying mechanisms. The findings reveal that patient capital can effectively alleviate financing constraints faced by upstream and downstream enterprises in supply chains through two pathways: accelerating the turnover speed of suppliers' accounts

receivable and increasing the credit loan ratio of customers.

This study contributes in two ways. First, it expands the research scope of the economic consequences of patient capital, extending the focus from individual enterprises to the supply chain level. Second, this research provides empirical evidence on the supply chain spillover effects of patient capital, thereby contributing to the understanding of how capital moves across different entities in supply chain.

2. Theoretical analysis and research hypotheses

The literature on patient capital has primarily evolved along two trajectories: one stream focuses on constructing measurement frameworks from either micro-enterprise or macro-level perspectives (Lin and Wang, 2017; Wu et al., 2022), while the other investigates its economic consequences, such as alleviating financing constraints and stimulating corporate innovation (Jiang and Wu, 2024; Yang and Tang, 2025). These studies provide a foundational understanding of patient capital's role and lay important groundwork for our further exploration of its transmission mechanisms within supply chains.

While existing literature offers support for patient capital's role in alleviating financing constraints, its potential spillover effects along the supply chain remain unclear. This paper posits that patient capital can generate such spillovers through two key mechanisms: the financing cost effect, whereby reduced capital costs for invested firms lessen their reliance on excessive trade credit from suppliers, and the credit expansion effect, whereby these firms act as informational bridges, lowering information asymmetry between financial institutions and their supply chain partners, thereby enhancing credit access for the latter. Consequently, we propose the core hypothesis:

H1: Patient capital investment generates supply chain spillover effects that significantly alleviate the financing constraints of major suppliers and customers.

3. Data and Characteristic Facts

Based on government-guided fund shareholding data, this paper calculates the patient capital investment ratio across various industries. As shown in Figure 1, this ratio exhibited a sustained upward trend from 2003 to 2023, with significant heterogeneity among industries, reflecting dynamic adjustments in key supported sectors aligned with developmental stages and policy priorities. To further examine supply chain spillover effects, we analyze the relationship between

patient capital and accounts receivable/payable along the supply chain. Figure 2 shows that before 2015, as patient capital investment increased, supplier accounts receivable decreased while customer accounts payable rose. This co-movement suggests a close association with improved working capital turnover in the supply chain, providing preliminary empirical evidence for spillover effects.

[Insert Fig. 1. Here]

[Insert Table 1 Here]

[Insert Fig. 2. Here]

4. Research design

4.1. Sample selection and data sources

Using data on top five suppliers and customers of China's A-share firms (2003 – 2023), we exclude entities under special treatment (e.g., ST/PT), those in finance and real estate, and samples with critical data gaps. The final sample includes 5,142 firm-supplier and 8,476 firm-customer links, compiled from CSMAR, CNRDS, and Wind.

4.2. Variable selection

1. Dependent Variable

Following Bian et al., (2021), this study employs industry-adjusted accounts receivable-to-sales and accounts payable-to-cost ratios to measure financing constraints for upstream suppliers and downstream customers, respectively. The specific formulas are as follows:

$$Rec_{i,t} = \ln \left(\frac{Receivables_{it}/Sales_{it}}{\frac{1}{I} \sum_{i \in I} (Receivables_{it}/Sales_{it})} \right) \#(1)$$

$$Pay_{i,t} = \ln \left(\frac{Payables_{it}/Cost_of_Sales_{it}}{\frac{1}{I} \sum_{i \in I} (Payables_{it}/Cost_of_Sales_{it})} \right) \#(2)$$

where the denominator represents the industry mean. If patient capital generates positive supply chain spillovers, the Rec of upstream firms is expected to decrease, while the Pay of downstream firms is expected to increase.

2. Independent Variable

We selected government-guided funds as a proxy variable for patient capital for two reasons: First, their institutional design embodies typical long-term orientation and risk tolerance characteristics, and they are explicitly recognized by authorities as patient capital. Second, their

investment activities constitute observable exogenous events with clearly measurable equity holdings, helping mitigate endogeneity issues. Thus, this variable provides a reliable empirical strategy for identifying supply chain spillover effects. Additionally, this paper employs relational debt and stability equity ratios as proxy variables for robustness testing.

3. Control Variables

The model also controls for the following variables: firm size (Size), leverage ratio (Lev), profitability (ROA), cash flow ratio (Cashflow), fixed assets (Fixed), revenue growth rate (Growth), proportion of independent directors (Indep) and ownership concentration (Top1). Variable definitions are reported in Table 2.

[Insert Table 2 Here]

4.3. Model specification

We design the following benchmark regression model to examine the supply chain spillover effects of patient capital:

$$Account_{j,t} = \alpha_0 + \alpha_1 patience_{i,t} + \delta Controls_{j,t} + \gamma_j + \lambda_t + \varepsilon_{i,t} \quad (3)$$

Where: $Account_{j,t}$ represents the accounts receivable (Rec) of primary supplier j to firm i in year t , or the accounts payable (Pay) of customer j to firm i in year t ; $patience_{i,t}$ denotes firm i 's investment in patience capital in year t ; $Controls_{i,t}$ represents control variables; γ_i and λ_t denote firm-specific and time-specific fixed effects, respectively; $\varepsilon_{i,t}$ is the random error term.

4.4. Descriptive statistics

Table 3 reports the descriptive statistics of the variables. For the dependent variables, the mean account receivable is -0.193, indicating that the receivables of upstream suppliers are below the industry average. In contrast, the mean account payable is 1.146, suggesting that the payables of downstream customers are significantly higher than the industry average. This provides preliminary evidence that patient capital has a more pronounced effect on alleviating financing constraints for downstream customers. The mean patient capital investment is 0.752%, reflecting a relatively low average shareholding ratio in the sample firms, which is consistent with existing literature (Yang and Tang, 2025). All other control variables fall within reasonable ranges.

[Insert Table 3 Here]

5. Empirical analysis

5.1. Preliminary Discussion

This section first examines the impact of patient capital on the focal firms themselves, establishing the premise for its subsequent transmission through the supply chain. The following regression model is specified:

$$Y_{i,t} = \alpha_0 + \alpha_1 \text{Patience}_{i,t} + \delta \text{Controls}_{i,t} + \gamma_i + \lambda_t + \varepsilon_{i,t} \#(4)$$

where $Y_{i,t}$ represents firm performance measures, namely interest expense, debt scale, and total factor productivity (TFP). The results in Table 4 show that patient capital significantly increases both interest expense and debt scale, indicating improved financing conditions and access to long-term capital. Furthermore, it also enhances TFP, suggesting better resource allocation efficiency. These improvements equip focal firms with the enhanced capacity to provide liquidity support to their upstream and downstream partners, thereby laying the mechanistic groundwork for positive spillover effects along the supply chain.

[Insert Table 4 Here]

5.2. Baseline Regression

Table 5 presents the supply chain spillover effects of patient capital investment. Columns (1)-(3) report the results for supplier accounts receivable. Column (1) shows the baseline estimate without controls or fixed effects, Column (2) adds firm and year fixed effects, and Column (3) further incorporates control variables. The coefficients on patient capital are -0.049, -0.023, and -0.022 in columns (1) to (3), respectively, all statistically significant at the 1% level. Columns (4)-(6) show consistent results for customer accounts payable. These findings collectively demonstrate that patient capital significantly reduces both supplier receivables and customer payables, confirming our hypothesis that patient capital generates supply chain spillover effects that alleviate financing constraints for upstream and downstream partners.

[Insert Table 5 Here]

5.3. Robustness tests

We conducted a series of robustness checks to ensure the validity of our baseline results. First, using stable equity and relational debt as alternative measures of patient capital yielded consistently positive results. Second, employing SA and FC financing constraint indices as proxy variables for financing constraints produced consistent findings. Third, regressions incorporating high-dimensional fixed effects and varying sample intervals yielded consistent results.

Furthermore, to mitigate potential endogeneity issues, we undertook the following steps: First,

we employed Heckman tests to address possible sample selection bias. Results indicate that our conclusions hold even after controlling for the potential impact of sample selection bias related to whether firms disclose specific information about major suppliers and customers. Second, we used the mean value of patient capital investment within the same industry and year as an instrumental variable to mitigate potential omitted variable bias. The results remained robust after passing the identification insufficiency test and the weak instrumental variable test. Finally, we employed propensity score matching to address potential unobservable omitted variable issues, conducting nearest neighbor matching, kernel matching, and radius matching, with results maintaining robustness. Detailed regression results for all robustness tests are presented in the appendix (Tables I, II, III, IV, V, and VI).

5.4. Mechanism tests

1. Financing Cost Effect

This study examines the financing cost mechanism through a mediation effect test. As shown in Table 11, patient capital investment significantly reduces the accounts receivable days of suppliers (Column 2), and this metric shows a significantly positive coefficient in the regression (Column 3), indicating that focal firms accelerate payments to suppliers, thereby alleviating upstream financing constraints. However, no significant change is observed in the accounts payable days of customers, suggesting that this mechanism primarily operates upstream in the supply chain, which aligns with existing research findings.

[Insert Table 6 Here]

2. Credit Expansion Effect

Given the information asymmetry prevalent in credit markets, patient capital serves as a key external financing source. Its long-term investment behavior sends a positive signal to the market, and this credit-enhancing effect may transmit along the supply chain, thereby improving the financing conditions for upstream and downstream firms.

We use the credit loan ratio as a proxy variable to conduct a mediation effect test. As shown in Figure 12, patient capital does not significantly affect the credit loan ratio of suppliers, indicating that the credit expansion mechanism does not hold upstream. For customers, however, patient capital not only significantly increases their credit loan ratio but also effectively alleviates their financing constraints, confirming that credit expansion serves as a valid transmission path

downstream.

[Insert Table 7 Here]

The findings above indicate that while patient capital investments can alleviate financing constraints for upstream and downstream enterprises in supply chains, the mechanisms through which they exert their effects differ. Specifically, patient capital eases financing constraints for supplier firms via its cost-of-capital effect, while it alleviates credit pressures for major clients through its credit expansion effect.

5.5. Heterogeneity analysis

1. Supply Chain Dependency

To examine the role of supply chain dependency, we grouped samples into high and low dependency categories based on the proportion of supplier purchases and customer sales. The regression results confirm that the spillover effects of patient capital are only significant in strongly interdependent relationships. This can be attributed to the fact that the benefits of maintaining stable partnerships outweigh the associated costs, motivating funded firms to provide liquidity support to their key partners.

[Insert Table 8 Here]

2. Financing Capacity

Based on the collateralizability of assets, we grouped the sample according to financing capacity to conduct heterogeneity tests. The results show that the supply chain spillover effects of patient capital are more pronounced among upstream and downstream partners with weaker financing capacity. This is primarily because firms with poorer financing capabilities face higher external funding needs, while core enterprises leverage their information advantages to identify and bear risks at lower costs. The infusion of patient capital further enhances this capacity for internal risk pricing and financing support.

[Insert Table 9 Here]

5.6. Further analysis

This study further examines the impact of patient capital on investment efficiency and innovation output of upstream and downstream firms in the supply chain. Using the Richardson model to measure investment efficiency (Richardson, 2006; Biddle et al., 2009), the results show that patient capital significantly reduces inefficient investment levels among supply chain firms

(Table 10). For innovation output, measured by the number of patent applications, regression analysis indicates that patient capital significantly enhances the innovation performance of upstream and downstream firms through supply chain channels (Table 11). These findings collectively demonstrate that by alleviating financing constraints, patient capital not only improves capital allocation efficiency but also fosters innovation activities within the supply chain, thereby providing critical support for supply chain modernization.

[Insert Table 10 Here]

[Insert Table 11 Here]

6. Conclusions

This study finds that patient capital significantly alleviates financing constraints for upstream and downstream firms in the supply chain through financing cost and credit expansion effects, with these spillover effects being more pronounced in contexts of higher supply chain dependency or weaker financing capacity among supply chain partners. Furthermore, mechanism tests confirm that patient capital enhances investment efficiency and innovation output in supply chain firms by easing financing constraints. These findings underscore the importance of patient capital in optimizing supply chain resource allocation and promoting industrial upgrading. Future research could extend to non-listed supply chain firms and explore the differential impacts of various types of patient capital.

CRedit authorship contribution statement

Xin Li: Methodology, Project administration, Data curation. **Xinlong Xiao:** Writing—original draft, Formal analysis.

References

- [1] Bian, Z. Y., Li, Z. Y., & Xu, M. Y. (2021). Development zone policy, supply chain participation, and corporate financing constraints. *Economic Research Journal*, *56*(10), 88 – 104.
- [2] Biddle, G. C., Hilary, G., & Verdi, R. S. (2009). How does financial reporting quality relate to investment efficiency? *Journal of Accounting and Economics*, *48*(2-3), 112 – 131.
- [3] Jiang, Z. Y., & Wu, F. X. (2024). Patient capital, digital economy, and innovation efficiency: Empirical evidence from A-share manufacturing listed companies. *Journal of Hohai University (Philosophy and Social Sciences)*, *26*(2), 121 – 133.
- [4] Lei, G. Y., & Liu, H. L. (2007). The nature of controlling shareholders, tunneling, and the degree of earnings management: Empirical evidence from the first loss reporting year of China's A-share companies. *China Industrial Economics*, (8), 90 – 97.
- [5] Li, W. A., & Ma, C. (2014). The "industrial + financial" model of industry-finance integration and corporate investment efficiency: Evidence from Chinese listed companies controlling financial institutions. *Journal of Financial Research*, (11), 109 – 126.
- [6] Lin, Y. F., & Wang, Y. (2017). New structural economics: Regarding "patient capital" as a comparative advantage. *Research on Developmental Finance*, *11*(1), 3 – 15.
- [7] Richardson, S. (2006). Over-investment of free cash flow. *Review of Accounting Studies*, *11*, 159 – 189.
- [8] Wu, M. J., Zhang, P., & Zhao, Z. Y. (2022). The impact of patient capital and innovation investment on firm performance: Evidence from SMEs listed on the SME Board. *Scientific Decision Making*, (9), 55 – 72.
- [9] Xiong, L. Y., Jiang, Y. M., Lian, L. S., et al. (2020). Controlling shareholders' leveraged share purchases and corporate cash holdings. *China Industrial Economics*, (8), 137 – 155.
- [10] Yang, G. Y., & Tang, L. (2025). The impact of patient capital on corporate resilience. *Finance & Economics*, (3), 15 – 29.
- [11] Zhang, X. Q., & Sun, L. J. (2017). Expanded collateral lists, over-leveraging, and corporate bankruptcy risk: The "double-edged sword" effect of chattel mortgage law reform. *China Industrial Economics*, (7), 175 – 192.
- [12] Zheng, W. P., & Gou, W. J. (2000). A study on the mechanism of industry-finance integration in China. *Economic Research Journal*, (3), 47 – 51.

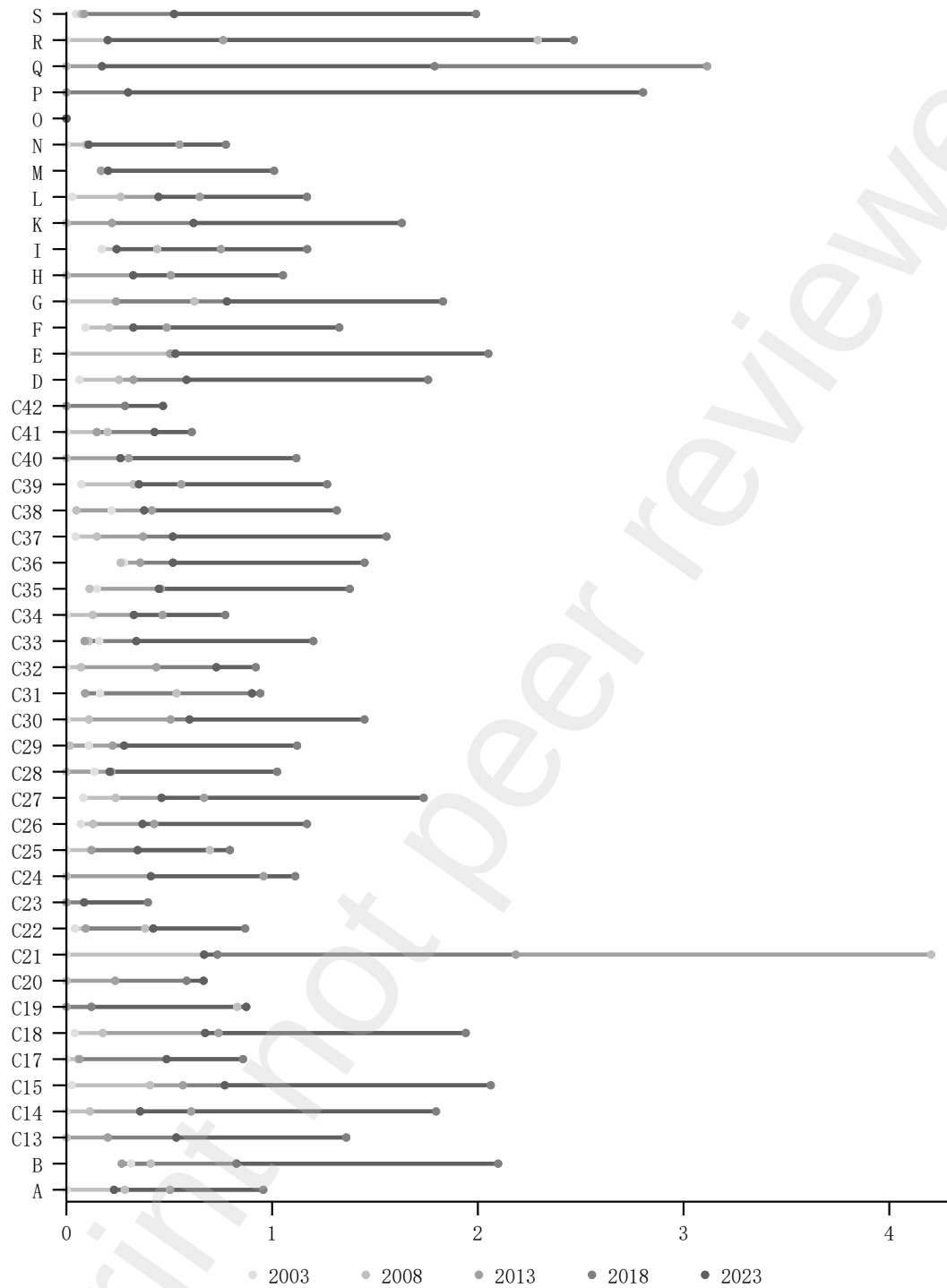


Fig. 1. Cross-Sectional Comparison of Industry-Specific Patient Capital Investment Ratios

Note: This graph uses the CSRC's 2012 industry classification codes. The manufacturing sector employs secondary-level codes, while all other industries use primary-level codes.

Table 1 List of industries

Industry code	Industry name
A	Agriculture, forestry, animal husbandry, fishery
B	Mining
C	Manufacturing
D	Electricity, heat, gas and water production and supply
E	Construction
F	Wholesale and retail trade
G	Transportation, warehousing and postal services
H	Accommodation and catering
I	Information transmission, software and information technology services
J	Finance
K	Real estate
L	Leasing and business services
M	Scientific research and technical services
N	Water conservancy, environment and public facilities management
O	Residential services, repairs and other services
P	Education
Q	Health and social work
R	Cultural, sports and entertainment
S	General

Note: This table presents the detailed information on the correspondence between industry codes and industry names for 19 sectors defined by the China Securities Regulatory Commission in 2012.

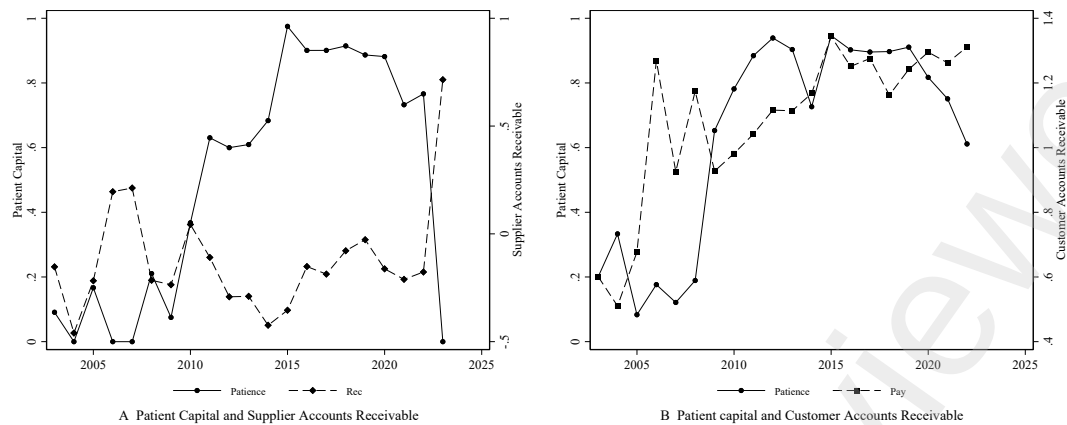


Fig. 2. Supply Chain Spillover Effects of Patient Capital

Note: The data in the graph is calculated based on matched samples, showing the average value of Patient Capital investments and the average accounts receivable (payable) of major suppliers (customers).

Table 2 Variable Definition

Type	Variable	Variable Definition
Dependent Variable	Rec	Proxy variable for accounts receivable from upstream suppliers in the supply chain.
	Pay	Proxy variable for accounts payable of downstream customers in the supply chain.
Independent Variable	Patience	Government-guided fund holding ratio.
	Patience2	Proportion of relational debt.
	Patience3	Proportion of stable equity.
Control Variables	Size	Natural logarithm of total assets.
	Lev	Total liabilities at year-end / Total assets.
	ROA	Net profit / Total assets.
	Cashflow	Net cash flow from operating activities / Total assets.
	Fixed	Ratio of fixed assets to total assets.
	Growth	Revenue growth rate.
	Indep	Proportion of independent directors to total board members.
	Top1	Shareholding ratio of the largest shareholder.

Table 3 Descriptive statistics

Variable	Obs	Mean	SD	Median	Min	Max
Rec	5142	-0.193	1.051	-0.063	-7.997	3.004
Pay	8476	1.146	1.038	0.952	0	9.564
Patience	13618	0.752	1.813	0	0	27.385
Size	13618	22.609	1.653	22.378	18.293	28.637
Lev	13618	0.464	0.210	0.471	0.010	1.282
ROA	13618	0.044	0.071	0.041	-1.859	0.695
Cashflow	13618	0.046	0.072	0.045	-0.496	0.539
Fixed	13618	0.225	0.166	0.188	0	0.869
Growth	13618	0.307	3.185	0.120	-0.985	140.241
Indep	13618	37.236	6.309	33.330	0	80.000
Top1	13618	0.364	0.162	0.336	0	0.900

Table 4 Preliminary Discussion

Variable	Interest Expense		Debt		TFP	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	2.032**	1.968*	0.155***	0.133***	1.736***	0.345***
	(1.995)	(1.937)	(9.812)	(8.655)	(9.424)	(2.667)
<i>Controls</i>	No	Yes	No	Yes	No	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	41337	41337	41337	41337	41337	41337
<i>Adj_R²</i>	0.066	0.066	0.815	0.824	0.803	0.889

Note: The values in parentheses represent robust t-statistics. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Fixed effects include firm-level and year-level fixed effects. The same applies to the table below.

Table 5 The Supply Chain Spillover Effects of Patient Capital

Variable	Supplier			Customer		
	<i>Rec</i>	<i>Rec</i>	<i>Rec</i>	<i>Pay</i>	<i>Pay</i>	<i>Pay</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	-0.049***	-0.023***	-0.022***	0.137***	0.125***	0.126***
	(-5.331)	(-3.314)	(-3.193)	(11.076)	(8.967)	(9.021)
<i>Size</i>			0.014			-0.010
			(0.371)			(-0.865)
<i>Lev</i>			0.383**			0.012
			(2.463)			(0.193)
<i>ROA</i>			-1.132***			-0.061
			(-4.216)			(-0.426)
<i>Cashflow</i>			-1.286***			0.130
			(-6.910)			(1.077)
<i>Fixed</i>			0.054			-0.207***
			(0.312)			(-3.045)
<i>Growth</i>			0.006***			0.000
			(8.390)			(0.332)
<i>Indep</i>			0.002			0.001
			(1.150)			(1.062)
<i>Top1</i>			0.229			0.106
			(1.217)			(1.537)
<i>Constant</i>	-0.152***	-0.186***	-0.775	1.050***	1.059***	1.225***
	(-9.402)	(-24.565)	(-0.900)	(86.980)	(106.307)	(5.206)
<i>Firm</i>	No	Yes	Yes	No	Yes	Yes
<i>Year</i>	No	Yes	Yes	No	Yes	Yes
<i>N</i>	5142	4781	4781	8476	7933	7933
<i>Adj-R²</i>	0.008	0.868	0.875	0.052	0.659	0.659

Table 6 Financing Cost Effect

Variable	Supplier			Customer		
	<i>Restrain</i>	<i>Day</i>	<i>Restrain</i>	<i>Restrain</i>	<i>Day</i>	<i>Restrain</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	-0.033***	-0.053***	-0.033***	-0.063***	-0.010	-0.063***
	(-10.215)	(-5.543)	(-10.095)	(-19.991)	(-1.085)	(-19.937)
<i>Day</i>			0.010*			0.006
			(1.791)			(1.256)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4410	4732	4400	7894	7874	7874
<i>Adj-R²</i>	0.583	0.461	0.584	0.584	0.406	0.584

Table 7 Credit Expansion Effect

Variable	Supplier			Customer		
	<i>Restrain</i>	<i>Credit</i>	<i>Restrain</i>	<i>Restrain</i>	<i>Credit</i>	<i>Restrain</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	-0.033***	0.001	-0.033***	-0.063***	0.013***	-0.067***
	(-10.215)	(0.503)	(-10.220)	(-19.991)	(4.684)	(-20.316)
<i>Credit</i>			-0.013			0.018**
			(-0.818)			(2.151)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4410	4781	4410	7894	6720	6681
<i>Adj-R²</i>	0.583	0.709	0.583	0.584	0.084	0.592

Table 8 Supply Chain Dependency

Variable	Supplier		Customer	
	High	Weak	High	Weak
	(1)	(2)	(3)	(4)
<i>Patience</i>	-0.029***	-0.009	0.176***	-0.002
	(-3.081)	(-1.157)	(8.811)	(-0.346)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	2409	2226	4080	3428
<i>Adj-R²</i>	0.882	0.902	0.685	0.672

Table 9 Financing Capacity

Variable	Supplier		Customer	
	High	Weak	High	Weak
	(1)	(2)	(3)	(4)
<i>Patience</i>	-0.007	-0.018*	-0.008	0.156***
	(-1.344)	(-1.682)	(-1.094)	(9.019)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	2238	2384	3275	4242
<i>Adj-R²</i>	0.908	0.898	0.690	0.667

Table 10 Investment Efficiency

Variable	Supplier		Customer	
	<i>Invest1</i>	<i>Invest2</i>	<i>Invest1</i>	<i>Invest2</i>
	(1)	(2)	(3)	(4)
<i>Patience</i>	-0.175***	-0.151***	-0.171***	-0.166***
	(-4.762)	(-3.213)	(-5.215)	(-4.136)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	4164	4354	6647	7060
<i>Adj-R²</i>	0.539	0.557	0.063	0.068

Table 11 Innovation Output

Variable	Supplier		Customer	
	<i>Innov1</i>	<i>Innov2</i>	<i>Innov1</i>	<i>Innov2</i>
	(1)	(2)	(3)	(4)
<i>Patience</i>	0.069***	0.088***	0.110***	0.121***
	(2.770)	(3.322)	(5.077)	(5.770)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	4525	4525	7933	7933
<i>Adj-R²</i>	0.902	0.872	0.084	0.093

Appendix

Table I Alternative definitions of variables

Variable	Supplier				Customer			
	Dependent		Independent		Dependent		Independent	
	SA	FC	Rec	Rec	SA	FC	Pay	Pay
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Patience</i>	-0.053***	-0.004***			-0.026***	-0.005***		
	(-18.975)	(-5.037)			(-9.472)	(-7.446)		
<i>Patience2</i>			-0.212***				0.191***	
			(-3.956)				(3.019)	
<i>Patience3</i>				-0.001**				0.006***
				(-2.295)				(6.961)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4781	4781	4715	3268	7605	7516	7599	5684
<i>Adj-R²</i>	0.947	0.902	0.880	0.872	0.955	0.897	0.617	0.656

Table II High-Dimensional Fixed Effects

Variable	Supplier		Customer	
	<i>Rec</i>	<i>Rec</i>	<i>Pay</i>	<i>Pay</i>
	(1)	(2)	(3)	(4)
<i>Patience</i>	-0.033***	-0.035***	0.134***	0.133***
	(-3.193)	(-3.956)	(10.129)	(10.498)
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>Industry</i>	Yes	No	Yes	No
<i>Industry-Year</i>	Yes	No	Yes	No
<i>Province</i>	No	Yes	No	Yes
<i>Province-Year</i>	No	Yes	No	Yes
<i>N</i>	5008	5074	8323	8332
<i>Adj-R²</i>	0.258	0.224	0.325	0.141

Table III Change sample interval

Variable	Supplier			Customer		
	<i>Rec</i>	<i>Rec</i>	<i>Rec</i>	<i>Pay</i>	<i>Pay</i>	<i>Pay</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	-0.043***	-0.023***	-0.022***	0.142***	0.125***	0.125***
	(-4.484)	(-3.437)	(-3.326)	(11.142)	(8.583)	(8.631)
<i>Constant</i>	-0.152***	-0.186***	-0.775	1.028***	1.041***	1.171***
	(-9.402)	(-24.565)	(-0.900)	(82.672)	(98.643)	(4.791)

<i>Controls</i>	No	No	Yes	No	No	Yes
<i>Firm</i>	No	Yes	Yes	No	Yes	Yes
<i>Year</i>	No	Yes	Yes	No	Yes	Yes
<i>N</i>	4083	3739	3739	7555	7036	7036
<i>Adj_R²</i>	0.007	0.866	0.873	0.059	0.654	0.655

Table IV Heckman Test

Variable	Supplier		Customer	
	First Step	Second Step	First Step	Second Step
	<i>Info</i>	<i>Rec</i>	<i>Info</i>	<i>Pay</i>
	(1)	(2)	(3)	(4)
<i>Patience</i>		-0.014*		0.142***
		(-1.955)		(7.185)
<i>Size</i>	-0.109	-0.028	-0.001	-0.023*
	(-0.446)	(-0.700)	(-0.045)	(-1.847)
<i>Lev</i>	-0.302	0.495**	0.210	0.054
	(-0.279)	(2.531)	(0.979)	(0.578)
<i>ROA</i>	-2.121	-1.340***	0.092	0.278
	(-1.074)	(-3.565)	(0.151)	(1.142)
<i>Cashflow</i>	-2.278*	-1.477***	-0.267	-0.024
	(-1.710)	(-6.018)	(-0.620)	(-0.118)
<i>Fixed</i>	3.163**	0.358	-0.080	-0.207**
	(2.334)	(1.336)	(-0.321)	(-2.401)
<i>Growth</i>	0.681***	0.003	-0.018	-0.019
	(3.298)	(0.972)	(-0.597)	(-1.638)
<i>Indep</i>	0.021	0.005*	0.007	0.005**
	(1.168)	(1.660)	(1.199)	(2.195)
<i>Top1</i>	-1.456	0.097	-0.772***	-0.126
	(-1.026)	(0.313)	(-3.301)	(-0.620)
<i>IMR</i>		0.133*		0.366
		(1.902)		(1.095)
<i>Constant</i>	0.365	-0.112	-1.231	0.795
	(0.060)	(-0.117)	(-1.423)	(1.562)
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	773	2935	3027	3027
<i>Adj_R²</i>		0.886		0.619

Table V IV Estimation

Variable	Supplier		Customer	
	First Step	Second Step	First Step	Second Step
	<i>Patience</i>	<i>Rec</i>	<i>Patience</i>	<i>Pay</i>

	(1)	(2)	(3)	(4)
<i>IV</i>	0.347***		0.842***	
	(4.486)		(12.575)	
<i>Patience</i>		-0.170***		0.062**
		(-2.863)		(2.014)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes
<i>N</i>	4700	4700	7817	7817
<i>Cragg-Donald Wald F</i>	92.601		498.828	
<i>Kleibergen-Paap rk LM</i>	42.041***		132.113***	

Tables VI Propensity score matching (PSM) results

Variable	Supplier			Customer		
	NNM	Kernel	Radius	NNM	Kernel	Radius
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patience</i>	-0.020**	-0.022***	-0.022***	0.145***	0.126***	0.126***
	(0.008)	(0.007)	(0.008)	(0.019)	(0.014)	(0.015)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Firm</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	2024	4712	3060	2608	7643	5592
<i>Adj-R²</i>	0.895	0.878	0.878	0.681	0.654	0.675